

RAM PRAKHYATH ANNAMAREDDY

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OBJECTIVE

Full stack engineer with hands-on experience building REST APIs, real-time collaborative systems, and AI-integrated applications across Java Spring Boot and Node.js.

EDUCATION

PES University, Bangalore, India [2023-Present]

BTech. – Computer Science & Engineering (CGPA : 7.94)

Relevant Coursework :

- Software Engineering
- Object Oriented Analysis and Design (Java)
- Cloud Computing
- Database Management Systems
- Big Data
- Operating Systems
- Computer Networks
- Data Structures and Algorithms

Certifications :

- Next.js 13: Your Odyssey into Full Stack Mastery – **Awarded Distinction Certificate.** [PESU-IO]
- OS Fundamentals and Linux – **Awarded Distinction Certificate.** [PESU-IO]

EXPERIENCE

Summer Intern - Centre of Data Modelling, Analytics and Visualization (CoDMAV), PES University [June 2025 - July 2025]

- CuraMate

[Tech Stack: React.js, FastAPI, PyTorch + scikit-learn (ML), Gemini (LLM)]

- Developed an AI-powered doctor portal to assist with medical triage, symptom clarification, and digital record management during patient consultations.
- Implemented PDF-based patient report parsing to extract demographic details and medical history, reducing manual data entry for doctors.
- Built ML-driven symptom analysis modules to categorize patient conditions and support faster clinical decision-making.
- Integrated LLM-powered clinical explanations and conversational assistance to provide doctors with concise, contextual insights.
- Designed backend APIs using FastAPI to store and retrieve patient records, enabling end-to-end digital workflows from appointment to prescription tracking.

SKILLS

Programming Languages	:	Java, Python, C, JavaScript
Databases	:	SQL (MySQL), Neo4j, PostgreSQL
Technologies	:	AWS, Linux (Arch, Debian), SSH
Frameworks & Libraries	:	Spring Boot, FastAPI, React.js, Next.js, Node.js, Express.js
Tools	:	Git, GitHub, Vim, Maven, Docker, Postman, Jira
Practices	:	Agile, Scrum, REST API Design, OOP, Design Patterns, Layered Architecture

1. Fortress: A Finance Data Backend Application with Role Based Access Control

[Tech Stack: Java, Spring Boot, Maven, SQLite, Postman (API testing)]

- Architected a layered REST backend (Controller–Service–Repository) in Spring Boot for financial data processing, with RESTful APIs supporting transaction filtering, partial updates, and categorization
- Implemented role-based access control (ADMIN, ANALYST, VIEWER) enforcing authorization at the service layer with SHA-256 credential hashing and basic authentication workflows
- Built analytics endpoints for income/expense aggregation, category breakdowns, and monthly trend reporting
- Designed robust exception handling and input validation with standardized HTTP responses across all endpoints
- Tested and documented all APIs using Postman, ensuring consistent request/response contracts

2. Homeserver: A Personal File Server on a Local Network

[Tech Stack: Java, Spring Boot, Maven, Docker, Watchtower, EndeavourOS, Linux]

- Designed and implemented a layered REST API backend (Controller–Service–Repository) in Spring Boot for file storage and retrieval over a local network, with endpoints for upload, download, and file management
- Containerized the application using Docker and automated deployments via Watchtower, enabling zero-touch updates by pushing new images to Docker Hub from the dev machine with no manual intervention
- Configured the server laptop to boot into TTY mode to minimize resource consumption, set up SSH for remote access and SCP for file transfers, and used systemd to auto-start Docker on boot ensuring containers restart automatically after power outages

3. Mditor: A Collaborative Markdown Editor

[Tech Stack: Next.js, Tiptap, Yjs, Node.js, Express.js, PostgreSQL, Docker, AWS EC2]

- Building a full stack collaborative Markdown editor with real-time multi-user sync using Yjs CRDT, currently migrating transport layer from y-webrtc to a self-hosted y-websocket server for reliability
- Developing a REST backend (Controller–Service–Repository) in Node.js/Express for document persistence and JWT-based authentication with bcrypt password hashing
- Designing PostgreSQL schema for user accounts and document storage with ownership and access control
- Deploying on AWS EC2 with Docker, with Next.js frontend and Express backend as separate containerized services

4. Hermes: A Local AI Assistant for Linux

[Tech Stack: Java, Ollama, REST API, Maven]

- Designing a local AI assistant that takes natural language input and executes Linux shell commands, running entirely on-device via Ollama with zero cloud dependency
- Architecting a clean separation of concerns across SystemContext, AIClient, CommandExecutor, and orchestration layers
- Building a system context engine that fingerprints the host environment (OS, kernel, shell, package manager, installed packages) to generate accurate, context-aware commands
- Implementing a safety confirmation layer requiring explicit user approval before any command is executed
- Designing a REPL loop for continuous interaction, maintaining system context across queries without repeated initialization

5. Efficient Knowledge Graph Retrieval for LLMs using Graph Partitioning and Hybrid Semantic Search

[Tech Stack: Python, Neo4j, PyTorch, Sentence Transformers, FAISS, NetworkX]

- Designing an efficient KG-RAG pipeline to improve LLM reasoning by combining semantic graph partitioning, ANN-based retrieval, and GNN inference
- Analyzing limitations of global GNN training on large knowledge graphs and proposing subgraph-level retrieval to reduce latency and noise
- Conducted an extensive literature survey on Louvain, Leiden partitioning, KG-RAG, GNN-RAG, and hybrid semantic search techniques
- Evaluated large-scale knowledge graphs (Wikidata-5M, DBpedia, ConceptNet) and justified dataset selection based on semantic richness and scalability
- Currently implementing a hybrid retrieval pipeline integrating graph embeddings, ANN search, and pre-trained GNN + LLM models

ACHIEVEMENTS & HACKATHONS

- Took part in **Hacknight 7 (2025)**, a hackathon held by **ACM PESUECC** in which **100+ students took part** and **placed 6th in the leaderboard**.
- **Grabbed the 2nd prize at Heal-O-Code 2.0**, an overnight hackathon held on 28th and 29th March 2025 by the **Centre for Data Modelling, Analytics and Visualization (CoDMAV) & WEAL of PES University, Electronic City Campus**.
- Took part in **Override 23'**, a hackathon held by **Google Developer Student Club (GDSC)** & **placed in the top 10 teams**.

PERSONAL DETAILS

Date of Birth : 25th August 2005
Languages Known : English, Kannada, Telugu and Hindi
Hobbies & Interest : Music (Vocals), Motorsports